

Paper Manager pack — DART (Nov 9 · Data)

Paper: DART: Noise Injection for Robust Imitation Learning

(Disturbances for Augmenting Robot Trajectories)

arXiv: <https://arxiv.org/abs/1703.09327> · [project](#) · [BAIR blog](#) · CoRL 2017 · PMLR 78:143–156

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PDF: </workspace/papers/dart-1703.09327.pdf> (= </workspace/dart-quiet/paper.pdf>)

Session: Nov 9 · due 9:30 AM · Type: **Data**

Role weights: Archaeologist=3 (LONGEST) · Stakeholder / Scientific Reviewer / Empiricist / Visionary also filled (seats otherwise empty)

Kanta role: unspecified — deepen lineage first; still produce **all five**

Themes: noise injection / recovery demos / covariate shift / safer than DAgger

Sources: Summarizer + Ablation Bot + Senior Research (</workspace/dart-quiet/DELIVERABLE.md>) + [paper_notes.md](#) / [env_outline/](#)

Sibling links: </workspace/droid-quiet/> · </workspace/act-aloha-quiet/> · </workspace/diffusion-policy-quiet/> · </workspace/oxe-quiet/> · </workspace/pi05-quiet/> · </workspace/openpi-comet-smoke/> · </workspace/behavior-1st-place-summary.md>

1. One-liner

Label recoveries by disturbing the expert, not by deploying the learner — protocol > architecture; heed ACT's warning that raw teleop noise can break fine manipulation.

Contrast: DAgger = on-policy corrections on *learner* states (unsafe / tedious); DART = *supervisor-side* optimized action noise so demos already contain boundary recoveries, while collection stays near the expert manifold.

2. Summary (exec overview + key numbers)

DART (CoRL 2017) is an **off-policy** Behavior Cloning variant that **injects optimized noise into the supervisor's control stream** during demo collection. The supervisor must demonstrate **recoveries** near the ideal-policy boundary; learning remains standard supervised BC on the aggregated noisy demos. Noise magnitude/shape is iteratively fit so the noisy supervisor approximates the **error of the trained robot** (Σ from residual outer-products, then α -shrinkage).

Why Data session: The durable artifact is a **demo-collection protocol** (noise schedule + α), not a new policy architecture.

Axis	Number / claim		
HSR clutter success (Fig. 3)	BC 49% \	DART($\alpha=3$) 79% \	DART($\alpha=6$) 72%
Relative lift vs BC	~+62% ((79-49)/49)		
Humanoid wall-clock	DART ~3× faster than DAgger		
Supervisor reward @ collect	DART -5% vs clean; DAgger ~-80%		
α story	Inverted-U — $\alpha=3$ best; $\alpha=6$ still > BC but hurts humans		
Isotropic $\Sigma=I$	Unsafe / weak — need optimized Σ		
MuJoCo domains	Walker, Hopper, Half-Cheetah, Humanoid; DART $\approx$ DAgger reward		
HSR protocol	4 humans; N=10 BC + 30/method; K=2; 20 evals		
Venue	CoRL 2017 · Berkeley AUTOLAB		

Action noise, not state noise (FLAG): Noise hits the **supervisor control** $\mathcal{N}(\mu)$ (Gaussian $\mathcal{N}(\pi^*(x), \Sigma)$) or discrete ε -greedy) → dynamics widen the state distribution. Distinct from obs dropout / domain rand / visual aug.

3. Keywords

DART; Disturbances for Augmenting Robot Trajectories; imitation learning; behavior cloning; covariate shift; compounding error; DAgger; noise injection; action noise; recovery demonstrations; data augmentation; supervisor-side noise; α -shrinkage; Σ^α ; MuJoCo locomotion; Toyota HSR; grasping in clutter; CoRL 2017; off-policy IL robustness; persistence of excitation; ACT chunking contrast; FM sample noise distinction; BEHAVIOR recovery demos

4. Deep dive — Motion / covariate shift / recovery control (PRIORITY)

Callout: This paper is a **primary syllabus node for covariate shift in IL**, not a classical motion planner. Control implications are direct and must be distinguished from later **chunking** and **FM sample noise**.

Three noises — keep them labeled

Layer	What it is	When applied	Example
(a) DART teleop / demo-collection noise	Optimized Gaussian (or ε -greedy) on supervisor actions during demos	Data collection	This paper; HSR $\alpha \in \{3, 6\}$
(b) FM / diffusion training / sample noise	Generative-head noise for score/flow matching	Train / inference sampling	π_0 / DP / BEHAVIOR'25 correlated FM
(c) Chunking / receding-horizon control	Predict action sequences; execute prefix; replan	Policy architecture	ACT (pred/exec chunks); DP obs=2/pred=16/action=8

Do not collapse (a)/(b)/(c). ACT cites DART [36] **CLEARLY**, then prefers **chunking** for fine teleop because “noise injection can directly lead to task failure.” DART attacks the *data distribution*; chunking attacks *effective horizon* — complementary, not substitutes.

Compounding error over horizon $\backslash(T)$

Same long-horizon pathology that later motivates ACT chunking, Diffusion Policy receding-horizon, and RTC soft-inpaint. Ross & Bagnell formalize the **Shift + Loss** decomposition; BC minimizes under supervisor dist; robot executes under its own → **compounding**.

Attack	Side	Mechanism
DART	Data	Widen supervisor traj so recoveries are labeled in-distribution
DAgger	Query / on-policy	Label learner states; visit low-reward states early
Chunking (ACT/DP)	Control	Shorten open-loop effective horizon; reactive replan
Keep-fails / scene churn (DROID)	Data ops	Coverage & recovery segments without live noise schedule

Recovery funnel (Fig. 1 moral)

1. Off-policy BC → drift off demo manifold → no recovery labels.
2. On-policy DAgger → learner visits dangerous states; human corrects under failure.
3. **DART** → small optimized disturbances → supervisor shows **boundary recoveries** while stay near high-reward manifold (“funnel”).
- Empiricist hook:** Script mid-traj kicks on Hopper; log recovery SR separately from nominal return. HSR clutter = qualitative long-horizon (clear → open-loop grasp) — course smoke uses locomotion recovery instead.

Vs DAgger (one-table)

	DAgger	DART
Who acts @ collect	Mixture / learner	Supervisor + action noise
Who labels	Supervisor on learner states	Supervisor’s own noisy demos
Retrain cadence	Every demo (full)	Infrequent (ψ update; final BC)
Dangerous states?	Often yes early	Mostly near expert funnel
Human load	High	Lower (small disturbances)
Humanoid collect reward	~~80%	~~5%
Humanoid compute	Baseline	~3× faster

5. Method deep dive — Σ^{α} , action vs state noise, iterative schedule

Problem (Eq. 1) → noise-injected BC (Eq. 2)

Minimize surrogate loss of learner under **its own** traj dist. BC instead fits under supervisor dist → Shift. DART collects from $\backslash(p(\backslash x \backslash \mid \pi_{\backslash \theta^*}, \backslash \psi) \backslash)$ and does ERM on that widened distribution.

Bound → KL objective (Lemma 4.1)

If $\backslash(\backslash \ln [0,1] \backslash)$, shift $\backslash(\backslash e^{-T} \sqrt{\backslash \frac{1}{2} D_{\backslash \mathrm{KL}}(p(\backslash x \backslash \mid \pi_{\backslash \theta^R}), \backslash p(\backslash x \backslash \mid \pi_{\backslash \theta^*}, \backslash \psi))} \backslash)$. Minimize KL ⇒ maximize likelihood of **learner’s actions** under noisy supervisor. Chicken-and-egg: final robot dist unknown at collection time.

Iterative fit (Eq. 3) + α -scaling (Eq. 4)

With current noise ψ_k and current learner $\pi_{\hat{\theta}}$: estimate ψ_{k+1} by MLE of robot residual under noisy expert demos, then shrink toward prior expected final error α .

Gaussian closed form (squared (L_2))

$$\begin{aligned} \hat{\Sigma}_{k+1} &= \frac{1}{T} \mathbb{E} \sum_t \big(\pi_{\hat{\theta}}(x_t) - \pi_{\theta^*}(x_t) \big)^2 \\ \hat{\Sigma}^{\alpha}_{k+1} &= \frac{\alpha}{T} \text{tr}(\hat{\Sigma}_{k+1}) \hat{\Sigma}_{k+1} \end{aligned}$$

- **Shape** from empirical error outer-products.
- **Magnitude** from α (expected simulated error $= \text{tr}(\Sigma)$).
- Estimate Σ on **held-out** prior-iteration demos (App §4.3).

Algorithm 1 checklist

1. Inputs: ψ_1^{α} , target α , iterations K , demos/round N .
2. For $k=1..K$: sample N demos under supervisor+noise $\rightarrow$ ERM on aggregate $\rightarrow$ update ψ then ψ^{α} .
3. Output: final θ^R by ERM on full aggregate.

Practical schedule knobs

Knob	Paper default
Noise family	Multivariate Gaussian on continuous $\mathcal{U}$ (ϵ -greedy in app.; unused empirically)
MuJoCo α	$\alpha = \text{tr}(\hat{\Sigma}_k)$ — “no extra prior”
HSR α	Multipliers 3 and 6 on $\text{tr}(\hat{\Sigma}_1)$ — 3 wins (inverted-U)
Isotropic baseline	$\Sigma = I$ always — fails
K	Small domains: update @ iters 2 & 8 ; Humanoid: 50 init then every 25 ; HSR $K=2$
Dagger β	0.5 (swept)
Data poverty	Subsample 50 (s,a)/traj (Ho & Ermon trick)
Architecture	Same 2×64 MLP as TRPO supervisor (MuJoCo); HSR = eye-in-hand CNN push + open-loop grasp

Moral: Wrong scale (isotropic, α too large for humans, Wishart $\text{Tr} \in \{0.005, 5\}$ extremes) hurts; magnitude often dominates structure *if* simulated error is right (App Fig. 5, $\text{Tr}=0.5$ can match DART).

6. Role-ready sections (all five)

6.1 Stakeholder

Product one-liner: A demo protocol that buys DAgger-level robustness **without** putting humans/robots through early catastrophic on-policy rollouts — **49% $\rightarrow$ 79%** on HSR clutter with moderate α ; Humanoid **3×** cheaper collect with only **−5%** supervisor reward drop.

Who pays / who benefits

- **Teleop / IL labs:** add optimized action-noise schedule to existing BC pipelines — **no new architecture**.
- **BEHAVIOR'26 entrants:** recovery-demo card is the missing data knob beside $\pi_{0.5}$ / N1.7 training noise.
- **Safety / IRB reviewers:** quieter collect-time noise beats on-policy visits to failed household states.
- **Non-consumers:** contact-rich fine teleop where ACT's warning bites — use **task-gated** noise (larger in loco/nav, smaller in precision insert) or prefer chunking.

Decision framing (Nov 9)

1. **Have a BC pipeline + teleop?** Inject **optimized** Σ^α (not isotropic vibes) after a small clean seed; log α + supervisor return@collect.
2. **Tempted by DAgger on humans?** Prefer DART-like off-policy recovery — paper never ran DAgger on physical HSR for a reason.
3. **Building 2026 data mix?** Budget a **recovery fraction** of demos with noise metadata (`recovery=true`).
4. **Fine manip?** Heed ACT — chunk first; DART noise as complementary coverage, not a substitute for horizon control.

Risks

- α is a **strong hyperparameter** (HSR 3 vs 6); humans degrade if noise too aggressive.
- Still iterative (not one-shot BC); needs held-out demos for Σ .
- 2017 HSR CNN stack $\neq$ modern VLA — steal the **idea**, not the codebase.
- Prop 4.1: if learner can fit expert perfectly, noise adds little.

Exam bite: “DART labels recoveries by disturbing the **expert**, matches DAgger reward with **safer/cheaper** collection (Humanoid $3\times$ / -5% vs -80%), and lifts HSR **49** $\rightarrow$ **79%** at $\alpha=3$ — protocol, not architecture.”

6.2 Scientific Reviewer

Claim under review: Optimized supervisor-side action noise yields DAgger-level IL robustness while staying off-policy, protecting collect-time reward/safety, and beating BC on human teleop.

Strengths

1. Clean BC vs DAgger positioning with a crisp **off-policy** alternative.
2. Theory (Lemma 4.1 + Prop 4.1) — KL story for why *some* Gaussian action noise beats pure BC when error > 0 .
3. Empirics span **algorithmic** (TRPO) and **human** supervisors.
4. Ablations vs isotropic & Wishart random Σ show **optimization of noise matters**.
5. Honest human result: larger α not always better (inverted-U).
6. Protocol transparency (Alg 1, closed forms, App §4.3 hyperparams).

Weaknesses / caveats

1. Lemma assumes bounded $\|\cdot\|_{\infty} \in [0,1]$; experiments use unnormalized L2 — bound is motivational.
2. Assumes supervisor **unaffected** by noise — paper's own $\alpha=6$ degradation is evidence humans violate this.
3. MuJoCo curves lack tabulated seed CIs; HSR $n=4$ supervisors, **20** evals — suggestive.
4. Half-Cheetah $\|\cdot\|_{\infty}=117$ vs common Gym ~ 17 — reproducibility caveat.
5. DAgger-B fairness: less frequent retrain hurts DAgger; modern online-update DAgger might close wall-time gap.
6. No DAgger on physical human task — HSR claim is vs BC only.
7. Data-poverty subsample (50 pairs/traj) may **amplify** shift vs modern large demos.
8. ϵ -greedy derived but unused empirically.

Verdict: Accept as **methodological CoRL’17 cornerstone** for recovery-as-data-augmentation. Idea aged better than the HSR stack. Suitable **Data** syllabus node; punish overclaim of Humanoid 3× / HSR 79% from Hopper Partial smokes.

Reviewer exam bite: Demand α sensitivity + isotropic control before accepting any “robust IL via noise” claim; cite Prop 4.1 assumptions and ACT’s fine-teleop warning as scope limits.

6.3 Empiricist

Hypotheses (from DELIVERABLE — feasible / falsifiable)

1. **H-dart>bc:** DART($\alpha\approx 1-3$) > BC on Hopper/Walker return (matched demos/arch).
2. **H-opt>iso:** Optimized Σ > isotropic $\Sigma=l$ at same intended scale.
3. **H-alpha-sweet:** α has interior optimum; too large drops collect-return / final policy (HSR moral).
4. **H-parity-dagger:** DART return $\approx$ DAgger; DART $\downarrow$ wall **or** $\uparrow$ supervisor return@collect.
5. **H-shift:** DART shrinks rollout–demo surrogate gap faster than BC (Fig. 4).
6. **H-recover:** Under scripted kicks, DART recovers $\geq$ BC.
7. **H-K:** K=2 after small BC seed captures most gains.
8. **H-offline-aug:** Residual-calibrated action-noise on DP/ACT demos = weak DART-lite (expect smaller lift).
- Falsifiers:** DART $\approx$ BC; isotropic $\geq$ DART; α monotone-good; no safety/wall gap vs DAgger; loss gap unchanged; kicks kill all equally; offline noise hurts SR.
- Minimal matrix:** {method: BC | isotropic | DART_ $\alpha\in\{1,3\}$ | (opt) DAgger_ $\beta=0.5$ } $\times$ {env: Hopper} $\times$ {N: 10–20} $\times$ {K: 2} + Tr(Σ) $\in\{0.005,0.5,5\}$ card.

Pri	Experiment	Paper anchor	Smoke	Est. GPU-h
P0	α magnitude + recovery vs BC	§5.2 Fig.3 · 49/79/72	Y (Hopper proxy)	1–6
P0	Optimized Σ vs isotropic $\Sigma=l$	Fig.2; App §4.3	Y	1–4
P0	Wishart Tr(Σ) sweep	App Fig.5	Y	2–6
P1	DART vs DAgger vs DAgger-B vs BC	Fig.2 · 3×/–5%/–80%	Y lite ≤ 20 iters	4–12
P1	Iteration + dual-loss (Fig.4)	App §4.3–4.4	Y (shared)	(shared)
P2	K schedule; offline DART-lite on DP/ACT	Alg 1; siblings	Partial	2–24
Defer / N	Humanoid-200; real HSR×4 humans; fresh TRPO	Fig.2–3	N	$\gg 40$ / IRB

Numbers to quote (canonical): HSR **49 / 79 / 72**; +62% rel; Humanoid **3× / –5% / –80%**; Tr=0.5 can match; isotropic fails.

Do not claim these from Hopper Partial.

Env outlines: /workspace/dart-quiet/env_outline/ (see §7).

6.4 Archaeologist — WEIGHT 3 / LONGEST

Role charge: Deepen prior $\rightarrow$ **DART** $\rightarrow$ **subsequent** lineage for IL robustness / covariate shift / recovery. Label **CLEAR** vs **TENTATIVE**. Cross-checked against local packs (ACT, DP, DROID, OXE, Octo, OpenVLA, π_0 , $\pi_{0.5}$).

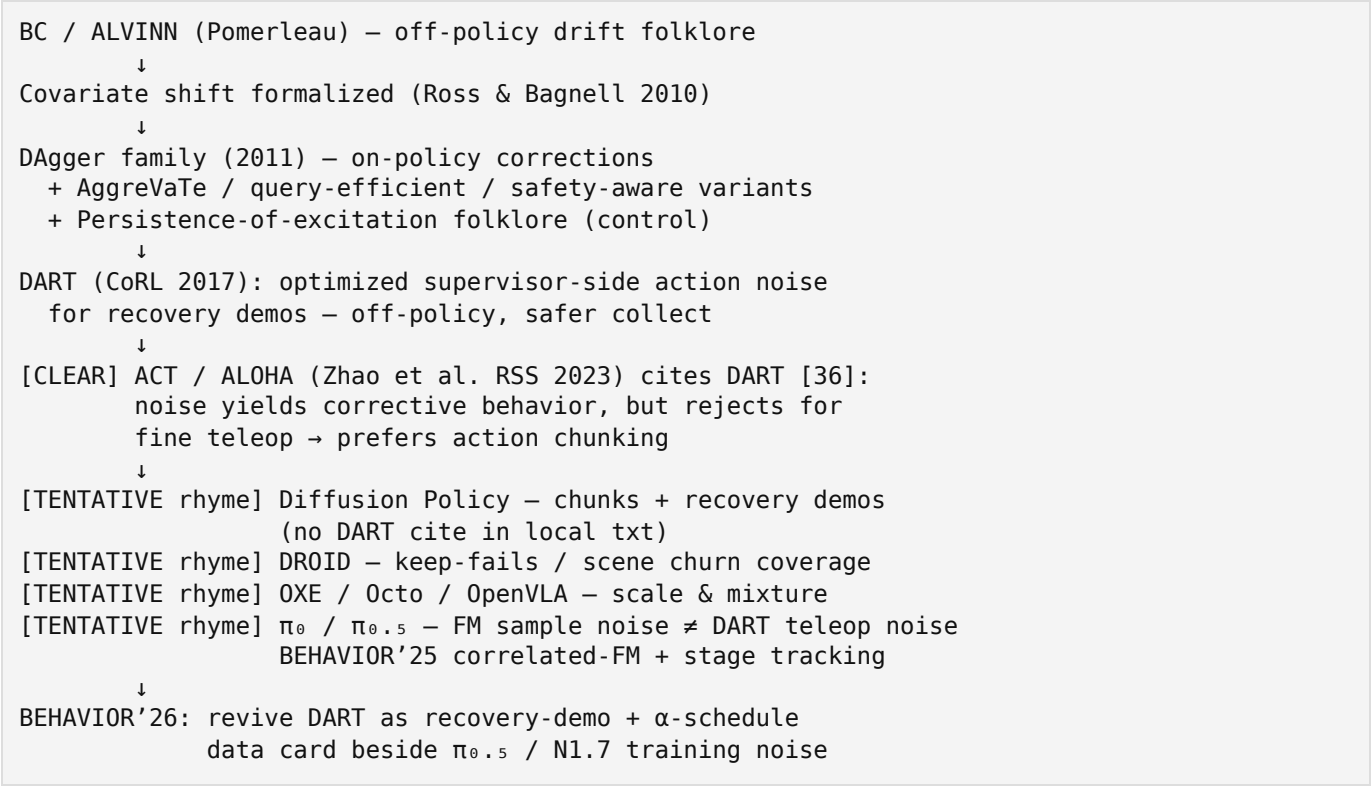
Priors (inputs this paper builds on)

Prior	Relation to DART	Confidence
Behavioral Cloning / ALVINN (Pomerleau 1989); Bojarski et al. self-driving BC	Classic off-policy supervised IL; road-edge drift = early compounding folklore	CLEAR (cited)
Covariate shift / compounding errors (Ross & Bagnell 2010)	Formalizes why BC fails under own-dist execution; DART's Shift term	CLEAR (cited [13])
DAGger (Ross, Gordon, Bagnell 2011)	On-policy iterative learner → supervisor queries; DART's main foil	CLEAR (cited [14])
SEARN (Daumé III, Langford, Marcu)	Intellectual ancestor of reduction-style IL; DAGger lineage	CLEAR as field prior; TENTATIVE as direct DART citation
AggreVaTe / Deeply AggreVaTeD (Sun et al. 2017 [20])	On-policy family siblings; cheaper on-policy updates	CLEAR (related work cites [20])
Query-efficient / safety-aware DAGger (Zhang & Cho 2016 [24]; Laskey et al. 2016 [7])	Document DAGger's human + safety + compute pain — motivates DART	CLEAR (cited)
Persistence of excitation / white-noise in adaptive/robust control (Green & Moore; Marmarelis; Sastry & Bodson)	Classical justification for rich control noise so models stay valid off nominal	CLEAR (cited §2.2)
Noise-injection teleop folklore (small Gaussian on expert actions)	Practice DART turns into an optimized algorithm with α -shrinkage	TENTATIVE as named prior; CLEAR that DART formalizes it
GAIL supervisor setup (Ho & Ermon 2016 [6])	Same TRPO deterministic-mean supervisor + 50-subsample protocol	CLEAR (cited)
Authors' prior: hierarchy-of-supervisors grasping (CASE 2016 [8]); human- vs robot-centric sampling [7]	AUTOLAB IL / grasping context for HSR	CLEAR

This paper's contribution (pivot)

- **Supervisor-side optimized action noise** as an **off-policy** antidote to covariate shift.
- Algorithmic: iterative **MLE noise fit** (Eq. 3) + **α -shrinkage** (Eq. 4), Gaussian / ϵ -greedy closed forms.
- Empirical: match DAGger robustness with **safer, cheaper** collection; beat BC on real human demos (49 → 79%).
- Syllabus slogan: “**Label recoveries by disturbing the expert, not by deploying the learner.**”

Genealogy



Subsequent lineage — CLEAR citations vs design rhyme

Work / practice	Relation	Confidence
ACT / ALOHA (2304.13705)	Cites DART [36] explicitly; noise → corrective behavior but “for fine manipulation... can directly lead to task failure”; motivates chunking	CLEAR citation
Diffusion Policy (2303.04137)	Compounding via chunks + receding-horizon; Push-T recovery demos. No DART cite in local txt	TENTATIVE design rhyme
DROID (2403.12945)	Diversity / scene churn / keep-failures for recovery research. No DART cite	TENTATIVE rhyme (coverage & recovery data)
OXE / RT-X (2310.08864)	Scale + mixture; failure-recovery upsampling lore — breadth not DART noise	TENTATIVE rhyme
Octo / OpenVLA / OpenVLA-OFT	Generalist BC/VLA; robustness from scale/co-train — not supervisor noise schedules	TENTATIVE (shared BC shift ancestor only)
π_0 / $\pi_{0.5}$	Continuous chunks + flow matching; BEHAVIOR forks add correlated FM noise (execution/sample noise, not DART teleop noise). No DART cite in local π txts	TENTATIVE rhyme; CLEAR distinction of noise layers
Gaussian action noise in modern teleop / OpenPI lore	Common practical knob; rarely attributed to DART	TENTATIVE folklore continuity
Domain rand / visual aug / obs noise	Orthogonal (perception) vs DART control-stream	CLEAR distinction (not descendants)
BEHAVIOR Challenge recovery bottlenecks	“No recovery demos” = failure mode; DART-style forced recovery labeling = plausible intervention	TENTATIVE application rhyme; high pedagogical value

Consistency vs prior local packs

- ACT pack: add **DART as the cited noise-injection prior** ACT explicitly discusses, then rejects for fine teleop.
- DP / π_0 / $\pi_{0.5}$: keep chunking & FM noise as **parallel anti-compounding tools**, not DART forks, unless CLEAR cite appears.
- DROID “keep failed demos” = **complementary data ops** to DART's *forced expert recoveries*.

Archaeologist exam bite: “Genealogy is **BC** → **Ross shift** → **Dagger** → **DART (optimized supervisor noise)**; ACT **CLEARLY** cites then prefers chunking for fine teleop; modern VLA recovery is mostly **TENTATIVE rhyme** (diversity, keep-fails, FM sample noise) — teach rhyme vs lineage, and never collapse DART teleop noise with FM training noise.”

6.5 Visionary — BEHAVIOR + follow-ups + apps (no Discord/Drive)

BEHAVIOR Challenge connection

- Long-horizon household tasks die from **unrecoverable mid-stage drift** (slip, miss grasp, occlusion) — exactly DART's boundary-recovery moral.
- 2025 winners leaned on **correlated FM noise inside the trainer** + System-2 stages — **not** Dagger human-query loops. DART is the missing **collection-time** knob.
- Two data levers: (a) **DART-style teleop action noise** so experts demonstrate restabilization; (b) retain/upsample **failure & intervention** segments (DROID / $\pi_{0.5}$ coaching).
- Prefer **task-gated** noise — larger in loco/nav/push, smaller in precision insert (heed ACT).
- Race on $\pi_{0.5}$ / **N1.7** substrates; treat DART as **data recipe**, not a 2017 MLP reimplement.

One narrative line: Dagger taught **query the robot's mistakes**; DART taught **make the supervisor rehearse the robot's likely mistakes**; 2026 should **schedule those rehearsals** into household demos beside FM training noise.

Follow-up research

1. Learn **state-dependent** heteroscedastic $\Sigma(x)$ instead of global Σ .
2. Combine **DART collection** with **chunked policies** (ACT/DP/ π_0) — noise for coverage, chunks for horizon.
3. Replace hand α with meta-learned target error from prior tasks.
4. VLA finetune: inject noise in **proprio/action** channels during co-train replay, not only pixels.
5. CLEAR three-way ablate: DART teleop noise vs **correlated FM sample noise** vs **RTC inpaint**.
6. Stage-aware recovery: stronger noise near stage boundaries / grasp commits (link `commit_s` + BEHAVIOR stage votes).
7. Publish a **BEHAVIOR recovery/noise card** (α schedule, Σ structure, per-stage flags, `recovery=true` metadata).

New applications

- Assistive teleop that stays high-reward under tremor.
- Autonomous driving lane-recovery demos.
- Surgical / robotic ultrasound where safe expert disturbances beat learner-in-the-loop.
- Sim-to-real locomotion where persistence-of-excitation meets IL.
- Household BEHAVIOR: slip/occlusion recovery teleop budget beside scene churn (DROID-like).

What not to do

- Blind isotropic action noise on dexterous bimanual demos.
- Claiming modern VLAs are “DART descendants” without citations — teach **rhyme vs lineage**.
- Reviving full Dagger on 50 household tasks for humans.

- Conflating FM noise, domain rand, and live recovery teleop in writeups.

Visionary exam bite: “For BEHAVIOR’26, revive DART as a **recovery-demo + α -schedule data card** beside $\pi_{0.5}/N1.7$ **training noise**; separate the three noises; steal protocol, not 2017 TF Humanoid numbers.”

7. Empiricist run pack (from DELIVERABLE)

Env setup

```
export DART_WORK_ROOT=${DART_WORK_ROOT:-$HOME/dart-smoke}
cd /workspace/dart-quiet # or copy env_outline to cluster
bash env_outline/setup_mamba.sh
# mamba activate dart-scout # name in conda_env.yaml
mkdir -p logs runs
# EDIT partition/account in slurm_*.sh
sbatch env_outline/slurm_hopper_smoke.sh # P0: BC vs isotropic vs DART( $\alpha=1$ )
sbatch env_outline/slurm_alpha_sweep.sh # P0:  $\alpha \in \{0.25, 1, 3, 6\}$ 
sbatch env_outline/slurm_tr_sigma_sweep.sh # P0:  $\text{Tr}(\Sigma)$  Wishart card
sbatch env_outline/slurm_dagger_compare.sh # P1 lite: DAgger vs DART
sbatch env_outline/slurm_recovery_kick.sh # recovery / covariate diagnostic
```

File	Role
env_outline/conda_env.yaml	mamba sketch (Torch + Gymnasium/MuJoCo + SB3 optional)
setup_mamba.sh	env create; MuJoCo EGL/OSMesa notes
slurm_hopper_smoke.sh	P0 BC vs isotropic vs DART($\alpha=1$) — Ablation #2 + #1 proxy
slurm_alpha_sweep.sh	P0 $\alpha \in \{0.25, 1, 3, 6\}$ — Ablation #1 proxy
slurm_tr_sigma_sweep.sh	P0 $\text{Tr}(\Sigma) \in \{0.005, 0.5, 5\}$ +DART — Ablation #3
slurm_dagger_compare.sh	P1 lite DAgger bake-off ≤ 20 iters — Ablation #4/#5
slurm_recovery_kick.sh	recovery kicks + Fig.4-style shift diagnostic — #6
run_checklist.md	GPU / TIME honesty preflight
README.md	cluster usage notes

Outlines only — not sbatch’d from quiet box. Prefer **frozen** PPO/SAC/MLP supervisor stand-in; do **not** retrain paper TRPO for smoke.

GPU / RAM / TIME honesty

Workload	VRAM	Host RAM	Disk	Wall	Smoke
Hopper BC / DART (≤ 20 demos, $K=2$)	0–8 GB (CPU OK)	8–16 GB	<5 GB	30–90 min	Y
α sweep $\times 4$	0–8 GB	16 GB	shared	2–6 h	Y
Dagger vs DART matched	0–8 GB	16 GB	shared	2–8 h	Y
Recovery-kick eval only	0–4 GB	8 GB	ckpts	minutes	Y
Walker / Half-Cheetah scale	0–8 GB	16–32 GB	<10 GB	4–16 h	Partial
Offline noise $\rightarrow$ DP/ACT FT	12–24 GB	32–64 GB	+ demos	8–24 h	Partial
Humanoid-200 paper protocol	CPU days / GPU hours	32 GB+	—	$\gg 1$ day	N
Real HSR human study	—	—	robot	—	N

Honesty rule: if `#SBATCH --gres=gpu:N` but no device → **fail**; log α , $\text{tr}(\Sigma)$, supervisor return@collect, #retrains, peak VRAM/CPU-time next to every return. Never claim Humanoid 3× or HSR 79% from Hopper Partial.

Success criteria vs paper

Paper claim	Feasible bar	Not required
DART ↓ covariate shift vs BC	Directional ↑ return and/or ↓ rollout-demo L2 on Hopper	Exact Fig. 2 curves
DART ≈ DAgger performance	Returns within noise on one loco env	Humanoid parity
DART cheaper / safer @ collect	↓ wall or ↑ supervisor collect-return vs DAgger	Exact 3× / −5% / −80%
Isotropic insufficient	DART > isotropic at matched scale	Paper “unsafe” qualitative
α matters	Interior α beats extremes	HSR 79% / 72%
Recovery demos help	Higher recovery SR under kicks	Real clutter push recoveries
Humans can use DART	Skip — note protocol only	4-supervisor HSR study

Pass: smoke card with **H-dart>bc** or **H-opt>iso** or **H-parity-dagger** (compute/safety) directional + α/Σ logs.

Fail / overclaim: “we reproduced CoRL Humanoid 3× / HSR 62%” without matching protocol/hardware.

Scout success

Working Hopper (or synthetic) card with **BC vs isotropic vs DART(α) and/or $\text{Tr}(\Sigma)$ sweep** directional returns + logged α / Σ trace / supervisor collect-return / wall time.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	3 · LONGEST (~14–16)	CLEAR vs TENTATIVE lineage; ACT cite-then-reject; three noises
Stakeholder	filled	Protocol product; 49 → 79; 3×/−5%/−80%; safer than DAgger
Reviewer	filled	Lemma assumptions; α inverted-U; isotropic control; HSR n=4
Empiricist	filled	Hopper α / Σ / Tr / DAgger-lite smokes + GPU honesty
Visionary	filled	BEHAVIOR recovery card; no Discord/Drive

Spine (paste into Slide Maker):

1. **Label recoveries by disturbing the expert, not by deploying the learner** — protocol > architecture.
2. **Key numbers:** HSR 49 / 79 / 72 (+62% rel); Humanoid 3× wall, collect −5% vs DAgger −80%; α inverted-U.
3. **Motion must-hits:** compounding over $\backslash(T)$; action-noise recovery funnel; **DART data-side vs chunking control-side**; distinguish DART teleop noise / FM sample noise / chunking.
4. **ACT CLEAR** cite then prefers chunking for fine teleop — complementary, not replacement.
5. **BEHAVIOR'26:** schedule recovery rehearsals into demos beside $\pi_{0.5}/N1.7$ training noise; Empiricist Hopper smokes validate schedule intuition only.

GitHub title sketch: [ECE 605] - DART Making Archaeologist (or seat if assigned)

Full Slide Maker brief: /workspace/dart-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/dart-quiet/KANTA_PACK.md
PDF	/workspace/dart-quiet/KANTA_PACK.pdf
Slide brief	/workspace/dart-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/dart-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/dart-1703.09327-summary.md
Ablations	/workspace/papers/dart-1703.09327-ablations.md
Paper notes	/workspace/dart-quiet/paper_notes.md
Env outlines	/workspace/dart-quiet/env_outline/
Paper PDF	/workspace/papers/dart-1703.09327.pdf

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